# Generic Template for Catalogue Columns and Flag Filters

This document provides a brief outline of the necessary columns to extract from the 2MASS PSC and XSC catalogues to create the smaller dataset used for cross-matching. It also provides more information on the "flag" index, additional columns to extract and combine into a quality flag value.

Each catalogue will need to be transformed from its current state in the DAC to separate files for ingestion into the cross-match process, to allow for a common interface for flexible matching across multiple different catalogues. A summary of this in its current form is provided below.

**NOTE**: 2MASS has a separate extended-source catalogue for the largest of its objects, resolved in images. Here we will be merging these observations into the main catalogue, replacing any from the point-source catalogue. To do so, start with the PSC and check for the `ext_key` flag being set; if it is, use the XSC entry instead of the PSC row. Supplement these with any XSC objects that do not have `pts_key` set, concatenating these to the ends of chunk tables as appropriate.

## ID

In both catalogues, the unique designation of the object is `designation`. However, to differentiate the source of the object, these strings should be prepended with "2MASS J" for PSC objects and "2MASX J" for XSC objects, respectively.

## Sky coordinates, uncertainty

For the PSC we use `ra` and `dec`, but for the XSC we want `sup_ra` and `sup_dec`, to take advantage of the increase in accuracy of the band-merged position. In the PSC, we wish to take the area-preserving geometric mean of the semi-major and semi-minor axes of the uncertainty ellipse,

$$\left(\prod_{i=1}^{n} \sigma_i\right)^{\left(\frac{1}{n}\right)},$$

or

```
sig_avg = sqrt(err_maj * err_min).
```

For the XSC, we do not have an obvious uncertainty to use. The documentation[1] states that the accuracy of the co-added centroid is approximately 0.3"; this is corroborated by a comparison with FIRST radio sources[2]. Hence we will use a constant 0.3" uncertainty in the 2MASS XSC objects.

## Magnitudes

For the PSC, use `[jhk]_m` for magnitude. `[jhk]` refers to each magnitude in turn, so e.g. `j_m` and `j_cmsig`, etc. For the XSC, as we want the most accurate possible total magnitudes, we require the Kron magnitudes, and hence use `[jhk]_m_k20fe` as our magnitude (`[jhk]_m` for short), and `[jhk]_flg_k20fe` as our "flag" column (represented below as the shorthand `[jhk]_flg` for symmetry with the PSC).

In both catalogues, the value should be replaced with a `NaN` if the object was not detected (either some kind of non-detection value in the original catalogue, or if the object is flagged as such below).

## Signal-to-noise ratios

For SNRs we can use the pre-computed values: `[jhk]_snr`.

## Best detection index

For 2MASS we use all three bands, J, H, and $K_s$. We prefer the $K_s$-band, then the J-band, then H. The following quality cuts apply, starting with a default flag value of `2` if none of these criteria are met.

For objects in the PSC, the flags are:

[1] 2MASS Documentation, II.3b

[2] 2MASS Documentation, II.3d5. Assuming FIRST observations have an RMS of 0.5" (Becker, White, & Helfand, 1995, ApJ, 450, 559) and the FWHM of the 2MASS-FIRST cross-match separations is ~1.4", and hence the standard deviation of the matches is ~0.6" (see Figure 3 of II.3d5; note this is smaller than the quoted RMS of the distribution, but this value is inflated by a background false-match rate), then we get a typical 2MASS uncertainty of $\sqrt{0.6"^2 - 0.5"^2} \approx 0.33"$.

| Flag | Flag Number | Criteria |
|---|---|---|
| Non-Detection | -1 | `bl_flg` == 0 OR `rd_flg` == 0 OR `rd_flg` == 9 OR `[jhk]_m` is NaN OR `[jhk]_cmsig` is NaN |
| Saturation | 0 | `rd_flg` == 3 |
| Low quality | 1 | `rd_flg` == 4 OR `rd_flg` == 6 OR `bl_flg` > 3 OR `cc_flg` != "0" OR `ph_qual` == "E" OR `[jhk]_cmsig` > 0.5 |

And for objects in the XSC, the flags are:

| Flag | Flag Number | Criteria |
|---|---|---|
| Non-Source | -999 | `cc_flg` != "0" AND `cc_flg` != "Z" |
| Non-Detection | -1 | `[jhk]_flg` == 9 OR `[jhk]_m` is NaN OR `[jhk]_msig` is NaN |
| Low quality | 1 | `[jhk]_flg` == 3 OR `[jhk]_flg` == 6 OR `[jhk]_msig` > 0.5 |

In the PSC case, `*_flg` and `ph_qual` are three-character strings, so the J-band `bl_flg` would be found via `bl_flg[0]` (or `bl_flg[1]` if one-indexed), and so forth. For XSC, each band has its own flag column, so should therefore be a single-character string accessed directly, while `cc_flg` in the XSC is now also a single-character string for the entire band-merged object, and hence applies to all three bands simultaneously if applicable.

If an object is flagged as `-1` in all three bands, or is flagged as `-999` at any point, it should be removed from the catalogue; for best detection, we consider first the $K_s$ band, then J, then H. If there are any $K_s$-band objects flagged as `2`, we accept them, otherwise we take the J-band if it is flagged `2`, then H. If no bands are, then we check the `1`-flagged sources (in the same order of priority), then `0`-flagged objects. As above, if there are no non-negative flags, we remove the object.

## Epoch and proper motion

In both catalogues, we keep the `jdate` column, recording the epoch of the observation for proper motion purposes, but since there are no proper motions for individual objects we only record the epochs, to allow for other catalogues to be moved to the 2MASS observation date.

## Zone overlap index

A flag for "is this the primary zone this object is in, or is this object part of the padding around this zone", required for all cases where chunking is performed to parallelise the cross-matching process.

## Reject catalogue

Any object that is removed from the final catalogue should be placed in a separate "reject" catalogue and provided separately. As these will fail to meet the best detection criterion that does not need storing; however, we still will require the zone overlap index.